

Descriptive Analytics Report: Customer Satisfaction

Best-Selling Dishes, Satisfaction Ratings, Ratings vs Votes, Dining vs Delivery Experience, Price Analysis, Value-for-Money, Engagement Density, Menu Breadth , Price Positioning, Rating Alignment , Cuisine Pricing , Delivery Review

Note on data:

This report uses a one-month *Zomato* dataset covering 5 restaurants (BrownBear , Crystal Restaurant & Bar , Doner King , Siddique Kabab Centre , Taco Bell) across 4 cuisines in Hyderabad, spanning 449 menu-item records. The dataset includes separate dining and delivery ratings, dining votes, menu items, item-level votes, and prices. It is used to perform descriptive analytics and identify patterns in customer satisfaction, menu popularity, and pricing during the study period. Delivery vote counts were not populated in this export (all zero), so engagement analysis below uses dining votes.

1. Objective

Build a descriptive analytics layer over Customer Satisfaction to answer twelve recurring questions:

- 1.** Which restaurants receive the highest customer satisfaction ratings?
 - 2.** Which menu items are the most popular among customers?
 - 3.** Is there a relationship between customer ratings and customer votes?
 - 4.** Where does the dining experience differ from the delivery experience?
 - 5.** How do menu prices vary across restaurants and cuisines?
 - 6.** Which restaurant offers the best value for money?
 - 7.** Which restaurant generates the strongest engagement per menu item?
 - 8.** Which restaurant carries the broadest menu, and does size drive votes?
 - 9.** Which restaurant is positioned as premium vs budget on pricing?
 - 10.** Which restaurant keeps dining and delivery experience most closely aligned?
 - 11.** Which restaurant's cuisine commands a pricing premium over the rest?
 - 12.** Which restaurant would benefit most from a delivery-service review?
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2. How This Would Work at Real Scale

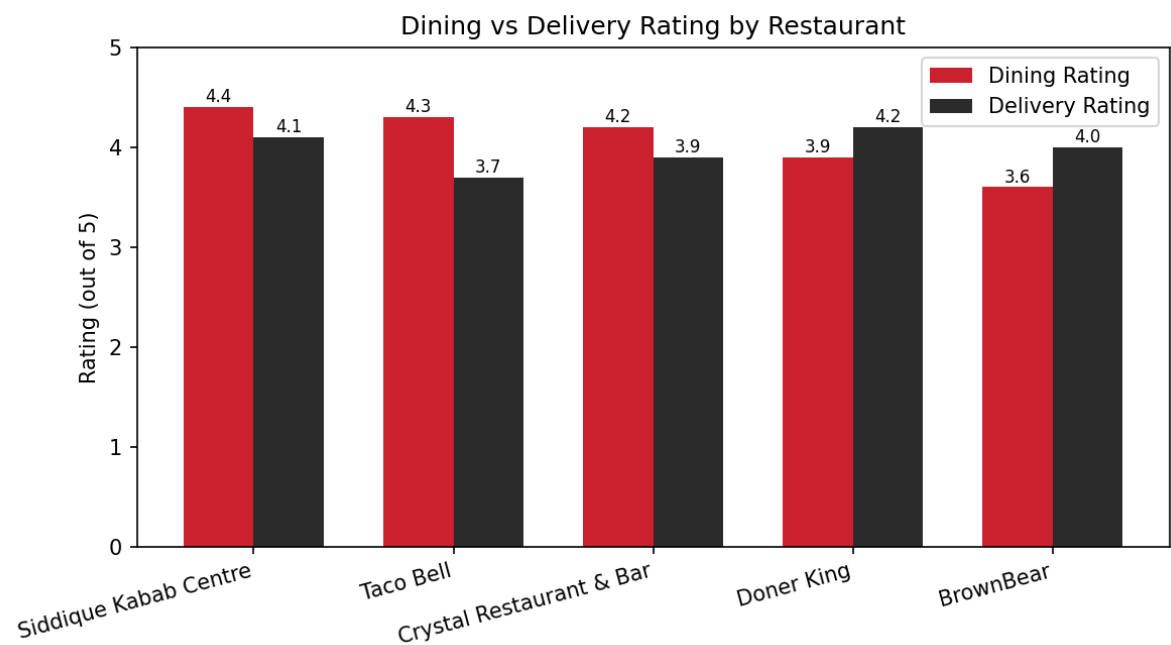
If customer ratings and restaurant interactions were collected in real time from millions of diners across cities, the analytical process would involve multiple stages:

Layer	Purpose	Typical tools
Investigation	Examine the dataset to understand its structure, coverage, and quality	Pandas
Cleaning & Standardization	Normalize restaurant names, cuisine labels, and price formats across sources	Pandas, Jupyter Notebook
Aggregation	Roll up item-level votes and prices into restaurant-level metrics	Jupyter Notebook
Analysis	Compute rating distributions, correlations, and dining-vs-delivery gaps	Pandas, Python, Jupyter Notebook
Visualization / Reporting	Publish rankings and trends to stakeholders on a recurring cadence	Jupyter Notebook , python

The report below reflects the output of that pipeline — restaurant- and item-level aggregated tables — not raw order-level data.

3. Restaurant Satisfaction: Dining vs Delivery

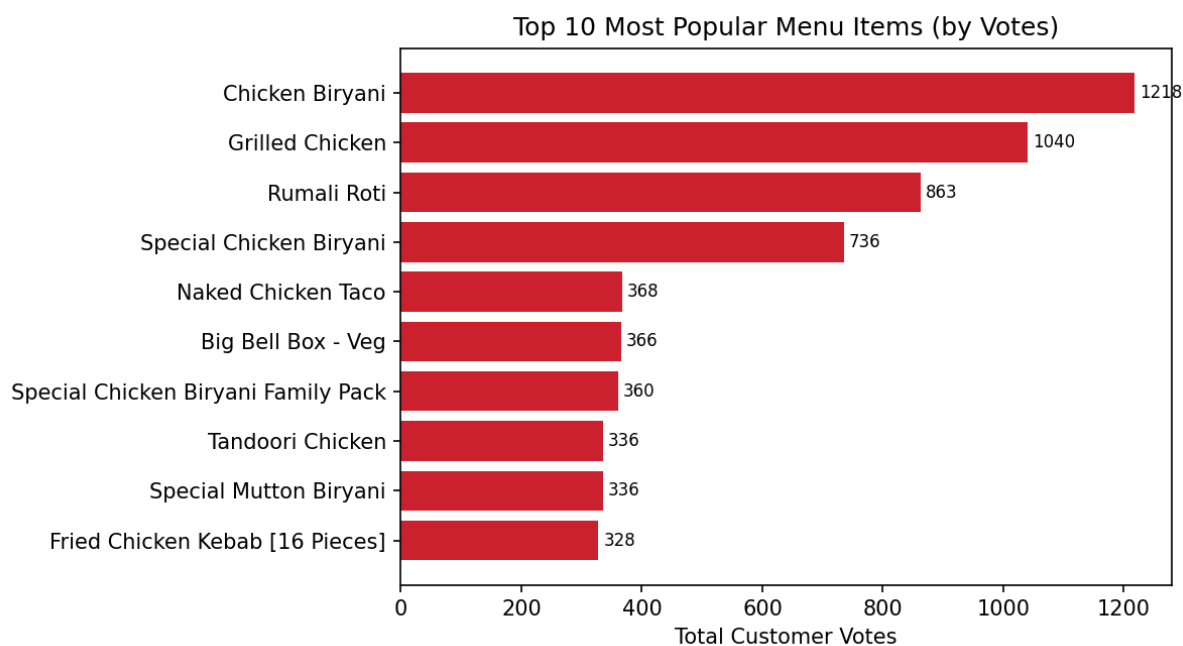
Each restaurant reports two separate satisfaction scores — a dining rating and a delivery rating — allowing the two experiences to be compared side by side.



Siddique Kabab Centre leads on dining experience (4.4/5), with **Taco Bell** (4.3) and **Crystal Restaurant & Bar** (4.2) close behind. On delivery, **Doner King** scores highest (4.2/5) despite the lowest dining score in the set (3.9) — the two rankings do not move together, which is explored further in Section 6.

4. Popular Menu Items

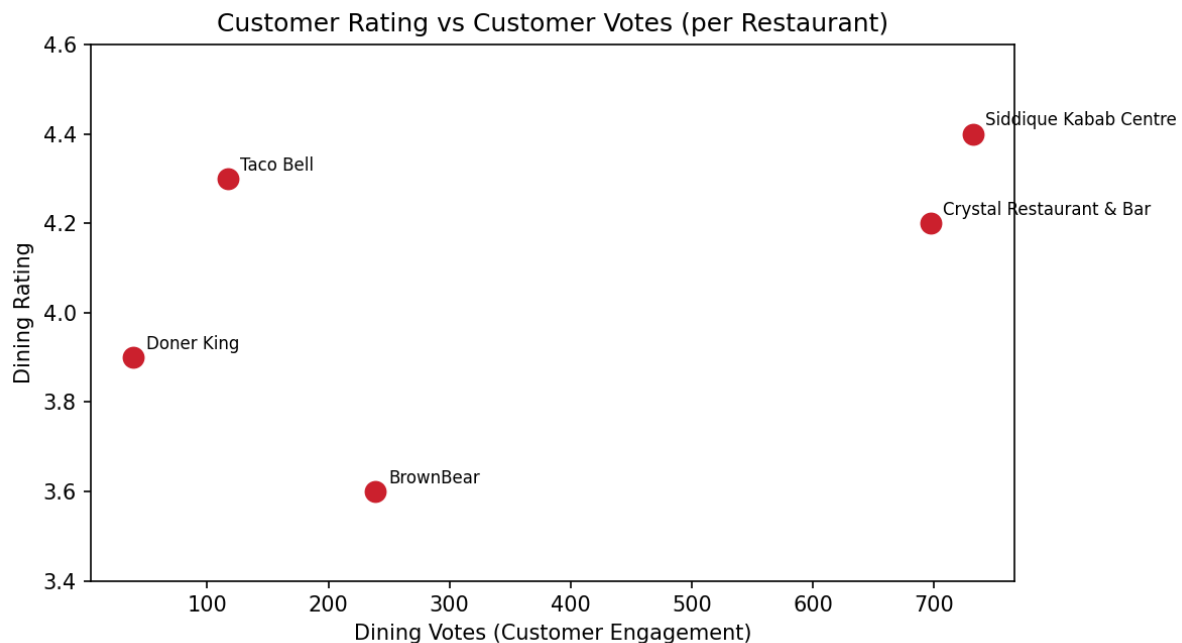
Menu items were ranked by total customer votes to identify the dishes customers engage with most.



Chicken Biryani , **BrownBear** is the most-voted dish (**1,218 votes**), ahead of **Grilled Chicken(Taco Bell)**(1,040) and **Rumali Roti(Siddique)** (863, spread across 2 restaurants). Chicken dishes take 7 of the top 10 spots — the clearest demand signal in the dataset — while biryani formats alone (Chicken Biryani, Special Chicken Biryani, Special Mutton Biryani, Family Pack) account for roughly a fifth of all top-10 votes.

5. Ratings vs Votes

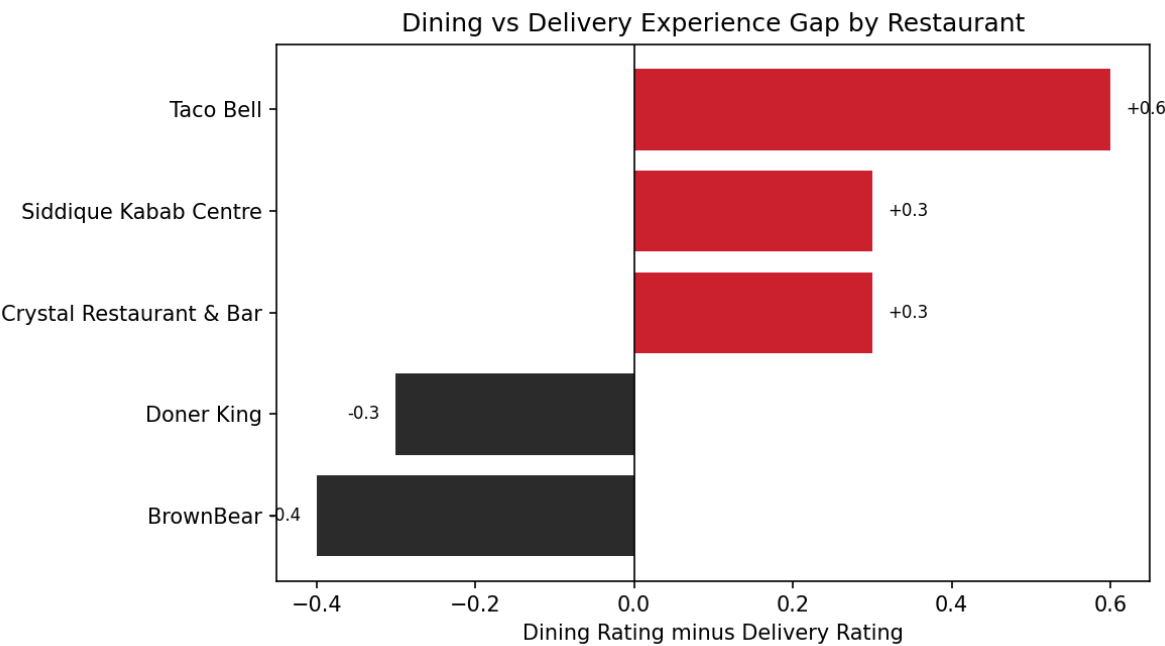
Dining rating was plotted against dining votes per restaurant to check whether higher engagement tracks with higher satisfaction.



Across these 5 restaurants the relationship is **positive but noisy** (correlation $\approx +0.52$): **Siddique Kabab Centre** and **Crystal Restaurant & Bar** combine high votes (732 and 697) with strong ratings, while **Taco Bell** earns a similarly high rating on far fewer votes (117) — with only 5 restaurants in this snapshot, this direction is suggestive rather than statistically firm and is worth re-checking as more locations are added.

6. Dining vs Delivery Experience Gap

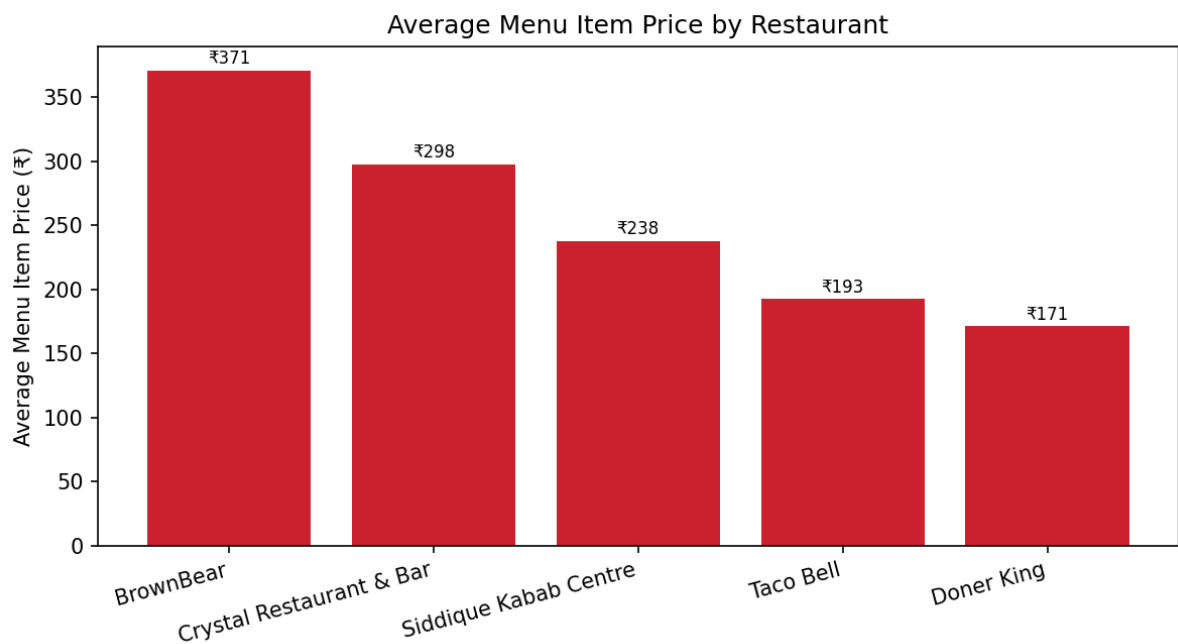
Subtracting delivery rating from dining rating for each restaurant highlights where the two channels diverge most.



Taco Bell shows the widest gap in the dataset (+0.6), with dining rated well above delivery (4.3 vs 3.7) — a candidate for a delivery-fulfillment review. **Siddique Kabab Centre** and **Crystal Restaurant & Bar** show smaller dining-favored gaps (+0.3). **BrownBear** and **Doner King** run the other way, rating better on delivery than dining (-0.4 and -0.3) — worth a closer look at in-restaurant service for those two.

7. Price Analysis

Menu item prices (**₹262 average, ₹249 median across all 449 items**) were compared across restaurants and cuisines.



BrownBear carries the highest average item price (**₹371**), followed by **Crystal Restaurant & Bar** (**₹298**). **Doner King** is the most affordable at **₹171** on average. By cuisine, **Biryani** commands the highest average price (**₹298**), while **Wraps** is the most budget-friendly category (**₹193**).

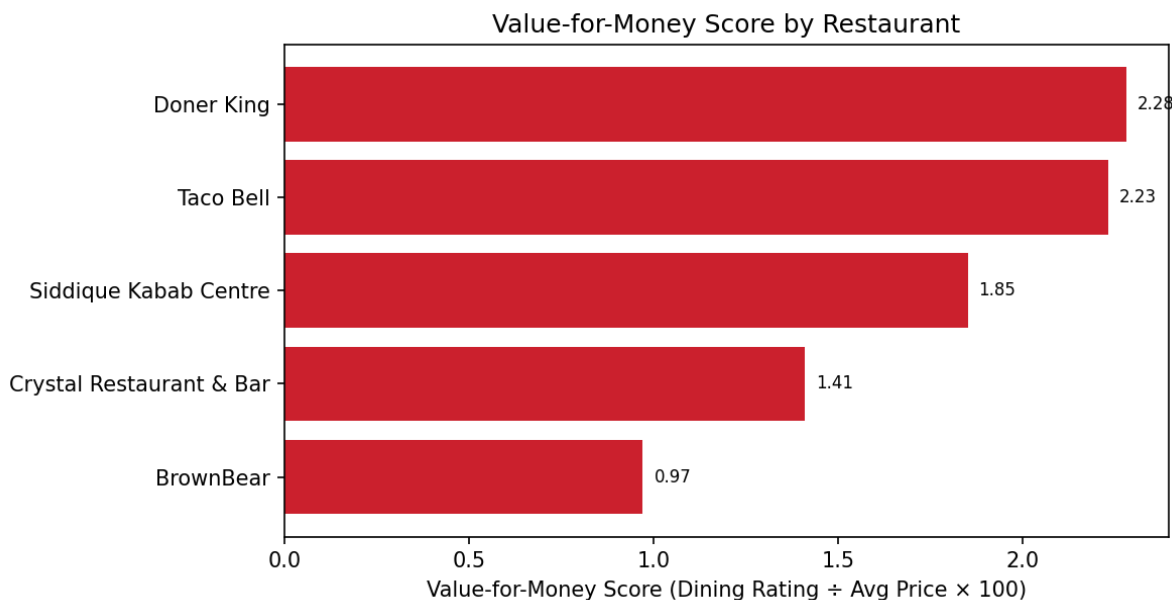
KPI

1. Value-for-Money Score ($\text{Rating} \div \text{Average Price}$)
2. Votes-per-Menu-Item (Engagement Density) = $\text{Total votes for a restaurant} \div \text{number of menu items}$

- Average item price (all restaurants): ₹262.40 (median ₹249)
 - Cheapest single item: Rumali Roti, Siddique Kabab Centre — ₹10
 - Most expensive single item: Chocolate KitKat Cake, BrownBear — ₹1,800
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8. Value-for-Money Score

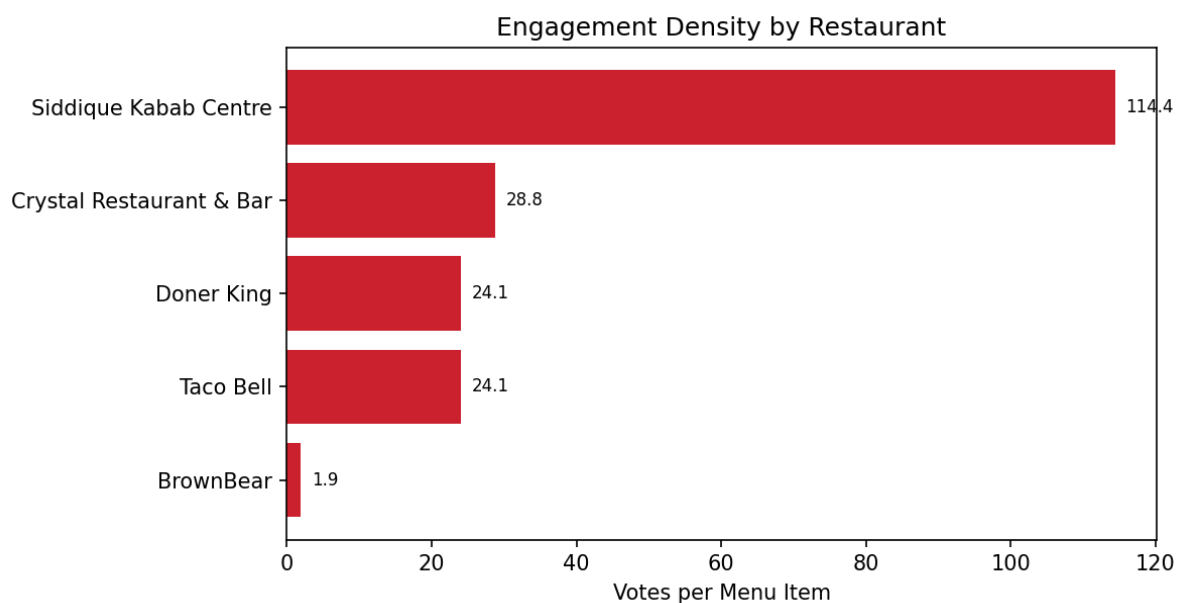
Objective 6 — Which restaurant offers the best value for money? Dividing dining rating by average item price surfaces which restaurant delivers the strongest rating per rupee spent.



Doner King serves this objective best, leading with a **2.28** value-for-money score — its low average price (₹171) more than offsets having the lowest dining rating in the set (3.9). **Taco Bell** is close behind at 2.23, earning its score from a strong rating (4.3) rather than a low price. **BrownBear** trails the group at 0.97, the combination of the lowest rating (3.6) and highest price (₹371) in the dataset.

9. Engagement Density

Objective 7 — Which restaurant generates the strongest engagement per menu item? Total item votes divided by menu item count shows how much attention each dish attracts on average.

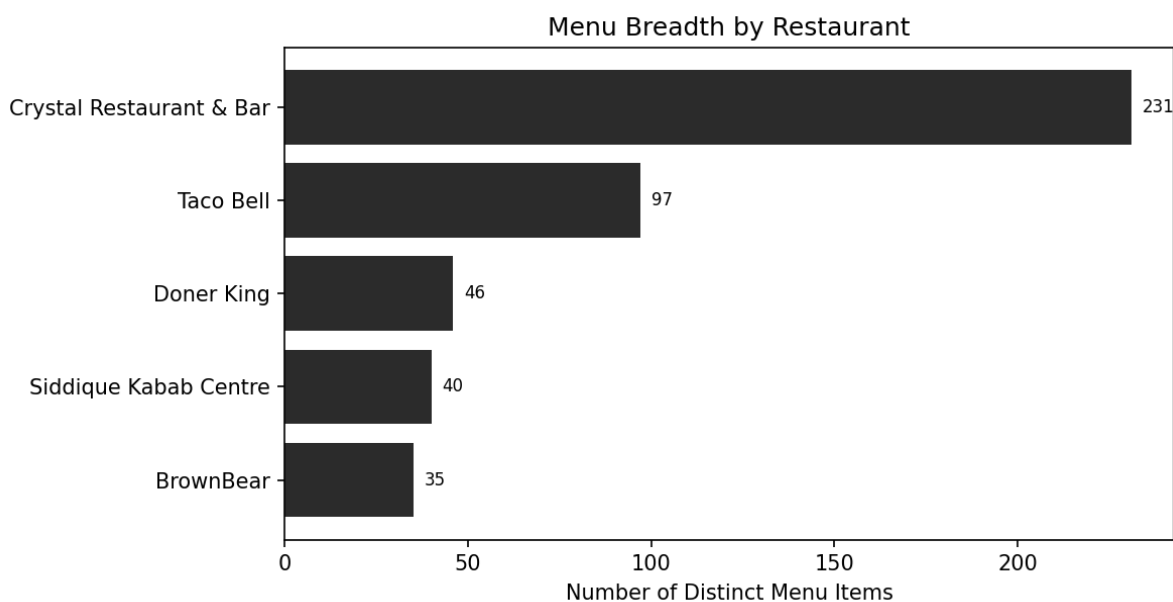


Siddique Kabab Centre serves this objective by a wide margin, at **114.4 votes per item** — more than 4x the next-closest restaurant — despite running a comparatively compact 40-item menu. **Crystal Restaurant & Bar** (28.8) and **Doner King** (24.1) follow at a distance, while **BrownBear** trails badly at 1.9 votes per item — its 35-item menu is barely generating engagement.

- Highest engagement density: Siddique Kabab Centre — 114.4 votes/item
 - Lowest engagement density: BrownBear — 1.9 votes/item
 - Average engagement density across restaurants: 38.7 votes/item
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10. Menu Breadth

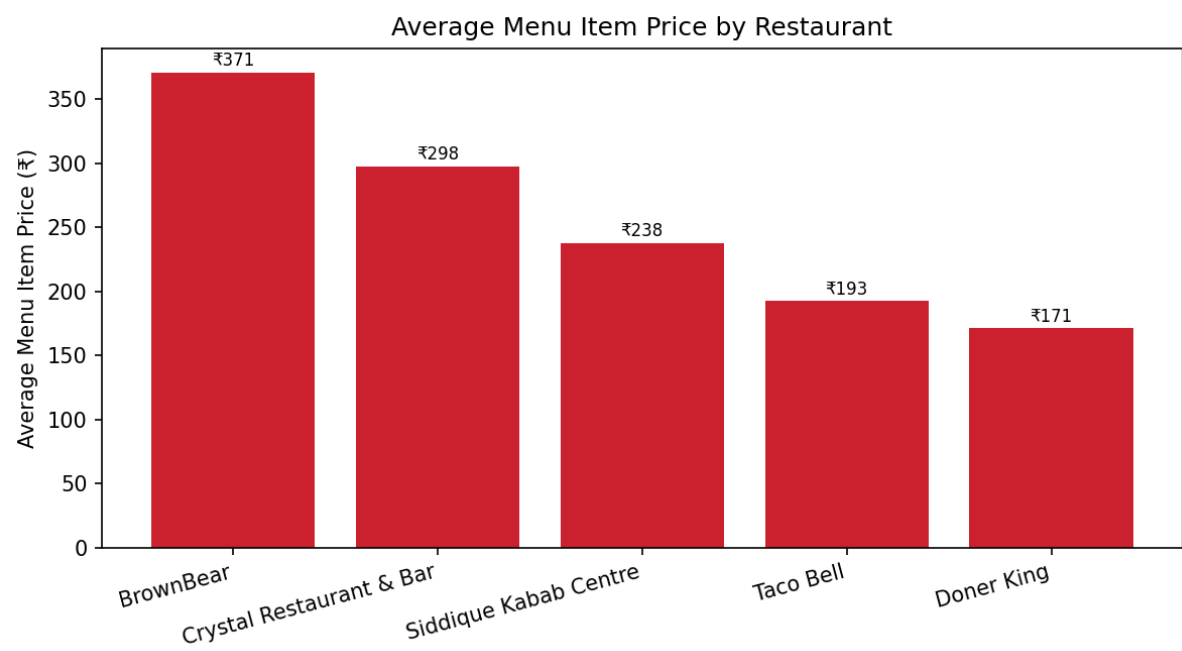
Objective 8 — Which restaurant carries the broadest menu, and does size drive votes? Comparing menu item counts against the engagement density above tests whether a bigger menu means more attention.



Crystal Restaurant & Bar serves this objective with by far the largest menu — **231 items**, roughly 2.4x the next-largest (**Taco Bell** at 97). Yet Crystal Restaurant & Bar converts that breadth into only moderate engagement density (28.8 votes/item, Section 9) — menu size alone does not drive votes; **Siddique Kabab Centre**'s much smaller 40-item menu proves that a focused menu can out-engage a large one.

11. Price Positioning: Premium vs Budget

Objective 9 — Which restaurant is positioned as premium vs budget on pricing? Reusing the restaurant-level pricing view from Section 7 to isolate the two extremes.

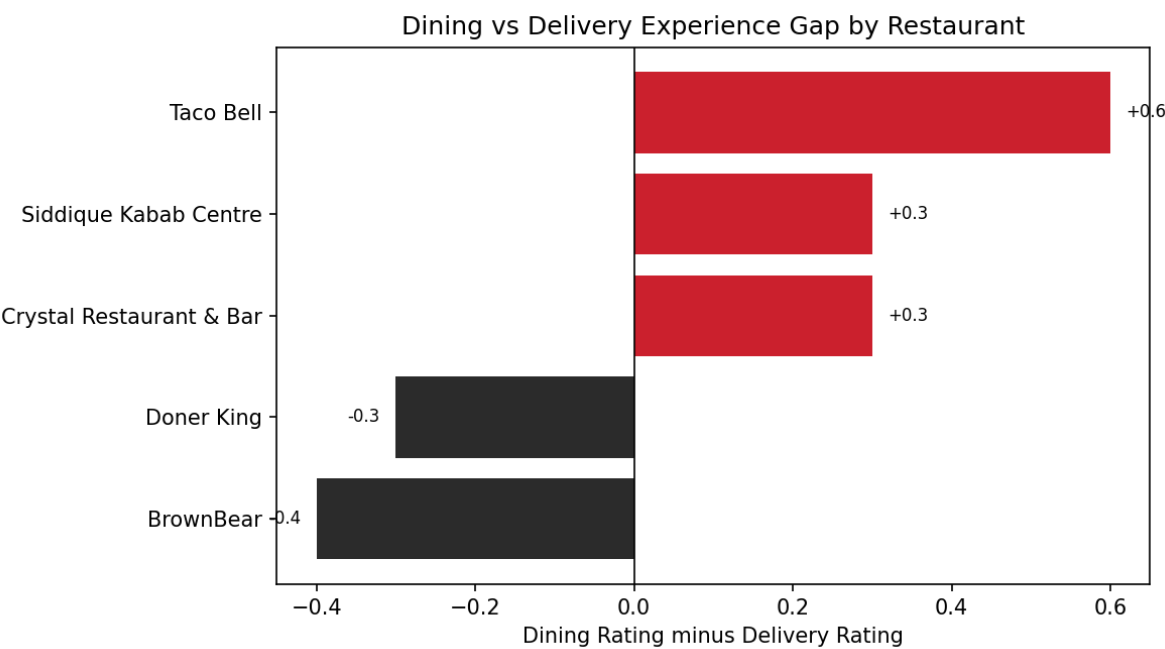


BrownBear serves the premium end of the market, pricing items at **₹371** on average — more than double the budget end. **Doner King** serves the budget end at **₹171** on average, the most affordable restaurant in the dataset.

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12. Dining–Delivery Alignment

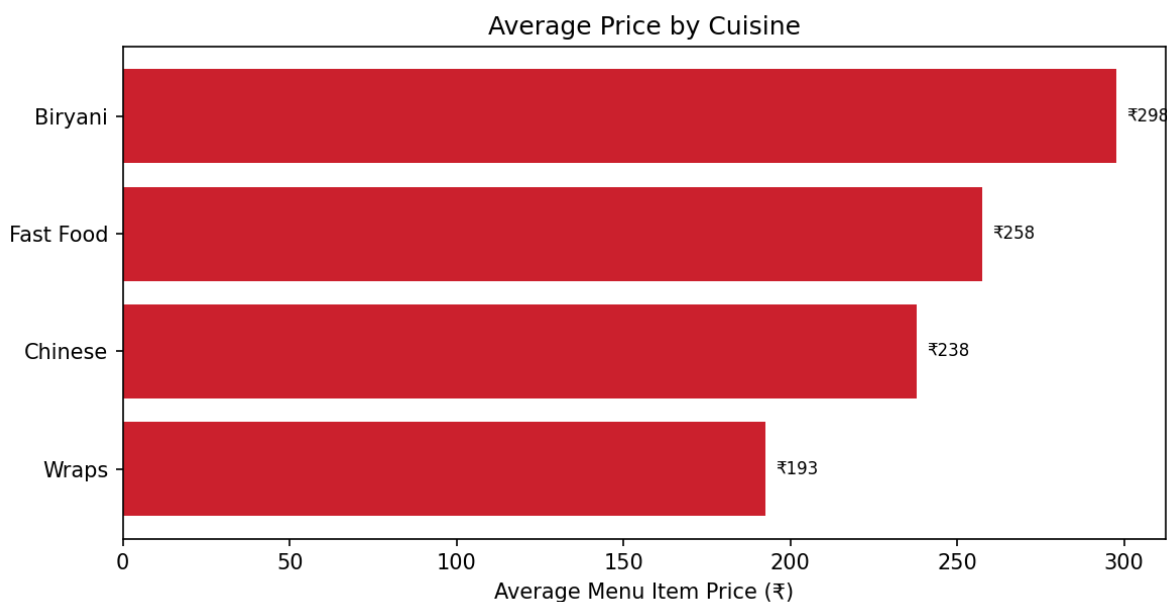
Objective 10 — Which restaurant keeps dining and delivery experience most closely aligned? Reusing the ratings-gap view from Section 6, this time reading the smallest bars rather than the largest.



Siddique Kabab Centre and **Crystal Restaurant & Bar** jointly serve this objective, each holding the smallest gap in the dataset at **0.3** — both restaurants keep dining and delivery ratings close, unlike the wider swings seen at Taco Bell, Doner King, and BrownBear.

13. Cuisine Pricing Premium

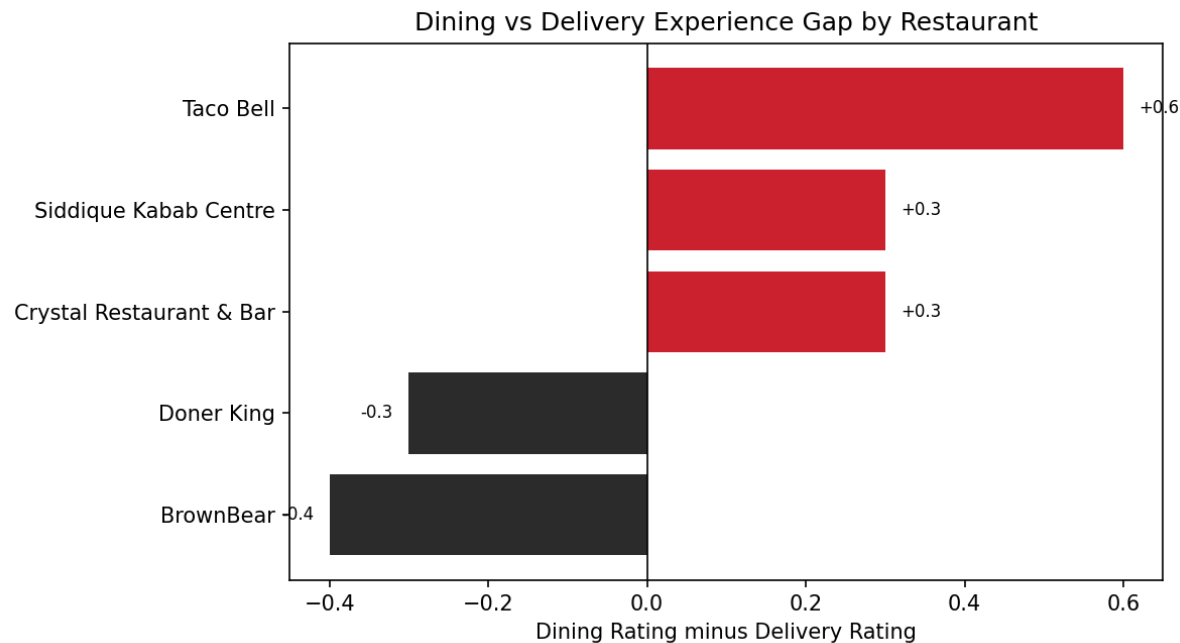
Objective 11 — Which restaurant's cuisine commands a pricing premium over the rest? Comparing average item price by cuisine category.



Crystal Restaurant & Bar serves this objective through its **Biryani** cuisine, which commands the highest average price (₹298) of any category in the dataset. **Taco Bell's Wraps** cuisine sits at the opposite end, the most budget-friendly category at ₹193.

14. Delivery-Service Review Priority

Objective 12 — Which restaurant would benefit most from a delivery-service review? Reusing the ratings-gap view once more, this time reading the largest positive bar — dining rated well above delivery.



Taco Bell serves this objective, showing the widest dining-over-delivery shortfall in the dataset at **+0.6** (4.3 dining vs 3.7 delivery). Its strong in-restaurant reputation is not carrying through to delivery orders, making it the clearest candidate for a delivery-fulfillment review among all five restaurants.

15. Summary of Key Metrics

Metric	Value
Restaurants covered	5
City covered	Hyderabad
Cuisine types covered	4
Menu items analyzed	449
Average dining rating	4.08 / 5
Average delivery rating	3.98 / 5
Top-rated (dining)	Siddique Kabab Centre — 4.4 / 5
Top-rated (delivery)	Doner King — 4.2 / 5
Rating–votes correlation (dining)	+0.52 (positive, small sample)
Widest dining-vs-delivery gap	Taco Bell — +0.6
Most popular menu item (by votes)	Chicken Biryani — 1,218 votes
Average menu item price	₹262.40 (median ₹249)
Best value-for-money	Doner King — 2.28 score
Strongest engagement density	Siddique Kabab Centre — 114.4 votes/item
Broadest menu	Crystal Restaurant & Bar — 231 items
Highest cuisine pricing	Biryani (Crystal Restaurant & Bar) — ₹298 avg

16. Next Steps to Move Descriptive → Predictive

This report answers what happened. The natural next layer, if useful:

1. **Delivery fulfillment review** for restaurants with the largest dining-vs-delivery gaps, before it shows up as churn
2. **Expanded engagement tracking** once delivery votes are populated, to compare dining and delivery engagement directly
3. **Multi-city rollout** of this same pipeline to test whether the rating–votes relationship holds beyond Hyderabad

Happy to build out any of these next, or refresh this report as new data becomes available — the same pipeline and chart structure above would apply directly.

THANK YOU